NOTE The cover page of this standard has been changed for administrative reasons. There are no other changes to this document.

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31 December 1979
SUPERSEDING
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18 January 1973

DEPARTMENT OF DEFENSE
MANUFACTURING PROCESS STANDARD

WELDING, HIGH HARDNESS ARMOR

AMSC N/A FSC THJM

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DEPARTMENT OF DEFENSE
Washington, DC 20301

Welding, High Hardness Armor
MIL-STD-1185(AT)

1. This Military Standard is approved for use by US Army Tank-Automotive Materiel Readiness Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Materiel Readiness Command, ATTN: DRSTA-GSS, Warren, Michigan 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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FOREWORD

This bookform standard covers welding of high hardness armor and contains instructions intended to standardize the preparation of specimens, welding, inspection procedures and the recording of data relating to armor procured to MIL-A-46100. This is not a procurement document.

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1. SCOPE

1.1 Coverage. This standard covers the fabrication of weldments constructed of 1 inch (25.4 mm) maximum thickness, high hardness, rolled armor and steel attachments welded to the armor. It contains information relating to preparation and submission of welding procedures, preparation of workmanship specimens, joint welding procedures, inspection of weldments and radiographic data.

1.2 Application. The following four processes are used in welding high hardness steel:

a. Shielded metal-arc (SMAW).
b. Gas shielded metal-arc (GMAW or GTAW).
c. Submerged-arc (SAW).
d. Flux cored electrode gas metal arc (FCAW).

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2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-6866	- Inspection, Penetrant Method of.
MIL-R-11468	- Radiographic Inspection, Soundness Requirements for Arc and Gas Welds in Steel.
MIL-M-11472	- Magnetic Particle Inspection, Process, for Ferromagnetic Materials.
MIL-M-11473	- Magnetic Particle Inspection, Soundness Requirements for Weldments.
MIL-E-13080	- Electrodes, Welding, Covered Austenitic Steel No. 19-9 Modified for Armor Application.
MIL-E-19822	- Electrodes, Welding, Bare, High-yield Steel.
MIL-E-19933	- Electrodes and Rods, Welding, Bare, Chromium Nickel Steels.
MIL-E-22200/1	- Electrodes, Welding, Mineral Covered, Iron Power, Low Hydrogen, Medium and High Tensile Steel, As-welded or Stress-relieved Weld Applications.
MIL-E-22200/2	- Electrodes, Welding, Covered (Austenitic Chromium-nickel Steel, for Corrosive and High Temperature Services).
MIL-E-22200/6	- Electrode, Welding, Mineral Covered, Low Hydrogen, Medium and High Tension Steel.
MIL-E-22749	- Electrodes and Fluxed, Submerged Arc-welding, High Yield Low Alloy Steels.
MIL-E-23765/2	- Electrodes and Rods-welding, Bare, Solid, Low Alloy Steel.
MIL-W-46086	- Welding of Homogeneous Armor by Metal-arc Processes.
MIL-A-46100	- Armor Plate, Steel, Wrought, High-hardness.

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STANDARDS

MILITARY

MIL-STD-453 - Inspection Radiographic.

USATARCOM

ATAC-STD-114 - Reference Standards and Radiographic Procedures for Partial Penetration Steel Welds.

PUBLICATIONS

29 CFR Part 1910 - OSHA General Industry Safety and Health Standards.

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY (AWS) STANDARDS

A2.4 - Symbols for Welding and Nondestructive Testing.
A3.0 - Welding Terms and Definitions.

(Application for copies should be addressed to American Welding Society, 2501 N.W. 7th Street, Miami, FL 33125.)

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3. DEFINITIONS

3.1 Contractor, manufacturer. The term "contractor" as used in this standard is defined as the organization having a direct contract with one Government activity. The term "manufacturer" is defined as the organization actually performing the operations covered by this standard. The contractor may, or may not, be the manufacturer.

3.2 Position drawing. The design agency will consider the need for radiographic, dye penetrant or magnetic particle inspection of weldments. When this need has been established and concurred in by Government engineering, a radiographic, dye penetrant or magnetic particle drawing (position chart) will be prepared by the design activity. These drawings will contain information similar to that contained in 5.5.1.

3.3 Radiographic inspection. Radiographic inspection is the use of x-rays or gamma-rays to detect discontinuities in material by presenting their images on a recording medium suitable for interpretation by qualified personnel.

3.4 Random positions. Random positions are those other than routine and are selected at the discretion of the Government.

3.5 Routine positions. Routine positions are those positions established as being in areas where weld defects are most likely to be encountered and are to be radiographed on each weldment selected for examination.

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4. GENERAL REQUIREMENTS

4.1 Welders.

4.1.1 Capability of welders. The contractor shall determine the capability of welding equipment or operators to perform welding, as prescribed in the contractor's or manufacturer's welding procedure, and meet the welding requirements established by workmanship specimens.

4.1.1.2 Welder's workmanship specimens. Workmanship specimens shall be a minimum of 12 X 6 inches (304.8 X 152.4 mm) and the same thickness as metal to be welded. The edge(s) to be welded shall be prepared by the same method(s) to be used in production welding. The edge subject to evaluation shall be the 12 inch (304.8 mm) length. Workmanship specimens are required for each different joint and welded attachment. Attachments which contain the same type weld may be grouped to reduce the number of workmanship specimens required. Grouping of attachments will be shown in the welding procedures.

4.2 Straightening. All weldments subjected to straightening shall have joints in the straightened area examined by a dye penetrant method or magnetic particle inspection process.

4.3 Ballistic plates. All thicknesses of high hardness armor up to 1/2 inch (12.7 mm) thick will require ballistic evaluation of weldments and preparation of 1/2 inch thick H-plates or corner-joints. The sizes for ballistic plates are specified in figures 14 and 15. The procuring activity shall determine the number and type of ballistic plates required. The number required will not be greater than ten.

4.4 Radiography.

4.4.1 Use of radiography. Radiography of production weldments under this standard is limited in amount and the provisions are intentionally stated in such a manner that normally no interference with production should result. As provided for, radiography is to be used as a tool for the development of satisfactory procedures and to provide sufficient spot checking at intervals so that a contractor will know when something has gone wrong with the procedure and can take steps to make the necessary corrections. Radiography as incorporated in this standard is not intended to be used for the purpose of ascertaining that every weldment accepted meets the specified standards. Instead, it is intended to assure that lowering of quality will be disclosed and that immediate remedial measures will be taken by the contractor.

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4.4.2 Number of radiographs. For the purpose of this standard, it is not advisable to require more than 40 radiograph films in a 24 hour day for each machine in use. The number of radiographs required for a particular vehicle, for routine spot checking as specified by the procuring activity, will not exceed the number which may be reasonably taken in an 8 hour day. Frequency of spot checking and the standard selected will be based upon the importance of the particular joint in the vehicle and upon the production schedule in the manufacturer's plant.

4.5 Hardness testing. For Brinell hardness testing, a 10 mm tungsten carbide ball shall be used.

4.6 Magnetic particle inspection. Magnetic particle inspection shall not be used on nonmagnetic weld filler metals.

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5. DETAIL REQUIREMENTS

5.1 Base metal. Base metal shall be in accordance with MIL-A-46100. Preparation of plate edges and subsequent joining by welding shall be performed by processes and procedures which minimize the reduction of hardness. On any ballistic surface, 5/8 inch (15.9 mm) from a plate edge, the Brinell hardness shall not be lower than that of the armor material.

5.1.1 Preparation of plate edges. The plate edges shall be prepared by grinding to alleviate any stress risers that may have been incurred in the cutting operation. Plate edges shall be checked for cracks using dye penetrant method conforming to MIL-I-6866, or magnetic particle process conforming to MIL-M-11472.

5.1.2 Preparation of surfaces for dye penetrant examination. Surfaces for dye penetrant examination shall be prepared by grinding. Blasted surfaces (sand, grit, and shot) are not acceptable.

5.1.3 Welded joints. On any ballistic surfaces 5/8 inch (15.9 mm) from toe of the weld, at any location of weldment, the Brinell hardness shall not be lower than that permitted by MIL-A-46100.

5.2 Procedures.

5.2.1 Preparation and submission of procedures. Detailed procedures listing all pertinent information shall be prepared for each different type edge preparation and for each different weld joint, including repairs. The recorded procedure information shall be prepared in triplicate on a form approved by the procuring activity. The cover sheet shall be signed by the manufacturer and the contractor, and shall be submitted to the procuring activity for review (see 5.4.1). For illustration in the preparation of a joint welding procedure, including armor and electrode types see figures 1 through 6.

5.2.2 Safety. The contractor/manufacturer shall assure that all procedures and work conform to 29 CFR part 1910.

5.3 Workmanship.

5.3.1 Workmanship specimens. Workmanship specimens shall be prepared by the manufacturer for each different edge preparation and weld joint, in accordance with recorded procedures, and subjected to the following evaluation. The edges and weld surfaces shall be examined by using the dye penetrant method, or magnetic particle process, in accordance with MIL-I-6866 or MIL-M-11472, as applicable. The specimens shall be free of cracks and other

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objectionable defects and shall represent the minimum acceptable pre-weld cleaning method and weld quality level to be used in production. A specimen 1 inch in width shall be removed from the workmanship specimens, etched to determine compliance with the welding symbol and hardness requirements, and attached to the workmanship specimens shown in figures 7 through 10. When these requirements have been met and the quality level of the workmanship specimens has been established between the contractor and the procuring activity, the welding procedure is considered approved for production application. Each specimen shall be identified by showing the date of approval, vehicle or component description and weld joint or plate preparation. Duplicate specimens may be required for use by the contractor at his facility.

5.4 Welding.

5.4.1 Joint welding procedure. Prior to production fabrication of any weldment, the contractor shall prepare, or have the manufacturer prepare, an isometric or perspective drawing showing the location of each joint and a joint welding procedure to cover all welding, including the repair of base material and welded joints. Each joint welding procedure shall contain a reference to the location of the joint, as shown in the isometric or perspective drawing, and detailed cross section sketch of the joint as specified in figure 6. It should also include the following factors, as may be applicable to the welding process selected:

a. Composition of armor and armor specification.
b. Brinell hardness range for all thickness.
c. Thickness of armor for each joint.
d. Welding symbol or symbols for each joint, as specified in ANSI/AWS A2.4.
e. Position in which welding will be performed.
f. Electrode identification.
g. Electrode sizes for all passes.
h. Welding current, arc voltage, and wire feed range.
i. Shielding gases and flow (cubic feet per hour).
j. Base metal temperature at start of welding, method of attaining temperature, and interpass temperature during welding.
k. Pass sequence for each joint.
l. Source of power, ac or dc, and polarity if dc is used.
m. Type of power source, variable voltage, constant potential, or other.
n. Description of process, as specified in AWS A3.0.
o. Use of spatter compound and identity of metal surfaces to be covered.
p. Method of process application; manual, semi-automatic, or automatic.
q. Method, or methods, of joint edge preparation.

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5.4.1.1 Filler metals. Austenitic or ferritic filler metals, conforming to specifications listed, with a minimum tensile strength of 80,000 psi shall be used. If filler metal for the process selected is not available in conformance with military specifications, the filler metal may be selected from AWS specifications. Choice of other filler metals must be approved by the procuring activity. When covered ferritic electrodes are used, they shall be of the low hydrogen type with moisture control. For electrodes not covered by a military specification, the chemical composition, yield strength, elongation, charpy V-notch, manufacturer and brand name will be shown in the joint welding procedure. For electrodes covered by a military specification; only the electrode specification, type or class, and brand name need be shown. Austenitic electrodes conforming to MIL-E-13080, MIL-E-19933 and MIL-E-22200/2; or ferritic electrodes conforming to MIL-E-19822, MIL-E-22200/1, MIL-E-22200/6, MIL-E-22749 and MIL-E-23765/2 shall be used, as applicable to welding method used.

5.4.2 Ballistic plate inspection. When a ballistic plate is required it shall be examined in such a manner as to assure that plates are acceptable for welding. After the plate is welded, it shall be tested at a Government proving ground. Testing and evaluation shall be accomplished in accordance with MIL-W-46086. The acceptance or rejection of test results shall demonstrate the suitability of fabrication procedures and will be the responsibility of the procuring activity. Welded armor data sheets (see figures 11, 12 and 13), shall accompany ballistic test plates to the proving ground.

5.5 Radiographic procedure.

5.5.1 Radiographic drawings. Radiographic drawings for review in establishing radiographic standards and "routine position" will be prepared and furnished by the design agency (see 3.2). Radiographic drawings will include the following:

a. An isometric or plan view (or both) of the weldment.
b. Identification, by letter or number, of the joint(s) to be radiographed as identified on the assembly drawing.
c. Identification of positions by letter or number.
d. Cross section of the joint(s) with welding symbol.
e. A table entitled "routine positions" (see 3.5) containing three columns titled; "position, left side"; "position, right side"; and "film size".
f. A table entitled "random position" (see 3.4) containing two columns titled, "positions" and "film size".

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(1) "Random shots" to be selected by the Government.
(2) All radiographs to meet standards of MIL-R-11468 or ATAC-STD-114.
(3) Radiography to be performed in accordance with MIL-STD-453.
(4) On consecutive exposures and weldments, when feasible, the direction of radiation shall be altered from parallel to the weld interface, to normal to the weld face, and to parallel to the opposite weld interface. The cycle shall be repeated for all positions radiographed on the weldment.
(5) Routine position, left side will be radiographed alternately with "routine position, right side" on consecutive weldments.

g. Nondestructive testing symbols will be shown as specified in ANSI/AWS A2.4.

5.6 Inspection.

5.6.1 Initial weldments and pilot vehicles.

5.6.1.1 Weldments. One-hundred percent dye penetrant method, or magnetic particle examination, shall be performed on welded joints in the first weldment produced. All cracks shall be rejected. If the first weldment is free of cracks, subsequent examination shall be performed on a random basis of at least 10 percent of total weld length on each subsequent weldment. If the first weldment is not free of cracks, the amount of examination on weldments shall be determined by the procuring activity. Random portable Brinell hardness checks shall be conducted on each completed weldment in 10 different locations to determine conformance to base metal hardness requirements (see 5.1) adjacent to weld areas. Each location checked shall be representative of the welded joint. The same amount of testing for hardness may be pro-rated against subassemblies as agreed between the contractor and procuring activity.

5.6.1.1.1 Repair of weldments. Dye penetrant method conforming to MIL-I-6866, or magnetic particle process conforming to MIL-M-11473 examination, shall be performed on all repairs to determine acceptance. Repair welding made on joints x-rayed in production will be subject to x-ray examination.

5.6.1.1.2 Radiographic requirements. Each initial weld joint to be subjected to direct ballistic attack shall be radiographed in accordance with the requirements of MIL-STD-453. Films for this weldment shall be submitted to the procuring activity with the radiographic position drawing (see 3.2) to which has been added following information:

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a. Location of film.
b. Direction of radiation.
c. Position of film (random or routine).
d. Stage of assembly.
e. Equipment used.
f. Film type.
g. Filter (size and location).
h. Focal film distance.
i. Film processing procedure.

With the additions shown herein the chart becomes a radiographic procedure and, in conjunction with the film of the initial production weldment, becomes the basis for establishing the radiographic frequency to be used in production.

5.6.1.2 Pilot vehicles. Radiographic examination shall be performed on all pilot vehicles, on all joints identified on the radiographic drawing, and shall comply with the designated standards.

5.6.2 Production weldments.

5.6.2.1 Welding procedures. All procedures used in the fabrication of weldments shall be subject to inspection for compliance with the recorded joint welding procedure.

5.6.2.2 Welds. All welds shall be subjected to visual inspection. Welds shall be equal to or better than workmanship specimens.

5.6.2.3 Straightening check. All plates with weldments shall be checked for straightness and, if straightening is required, the weldments shall subsequently be examined as specified in 4.2.

5.6.2.4 Weld repairs marking check. All repairs made to weldments shall be identified by suitable legible markings, of such nature that marking shall not be obliterated in handling. Marking shall be subject to Government approval.

5.6.3 Radiographic inspection. Less than 100 percent inspection may be permitted when an adequate statistical quality control system has been established by the contractor or manufacturer and approved by the Government.

5.6.3.1 Spot checking. Joints subject to direct ballistic attack shall be spot checked by radiography and other means to be established by the procuring activity. These other means, number of exposures, position and radiographic standards will be determined after examination of film and other information submitted.

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5.6.3.2 Joint basis. Normally, one film will be required on any specified joint in accordance with the established frequency determined by the relative importance of the joint in the weldment. Thus, spot checking will be on a joint basis and will not ordinarily require radiographing all joints in any one weldment.

5.6.3.3 Daily exposure. The number of radiographic films to be made daily shall not exceed the number required by the contract. The contractor may elect to make additional films.

5.6.3.4 Rejectable joints. When the radiographic spot check of a weldment indicates a rejectable defect in a particular joint, the remainder of the joint shall be radiographed. All defects found in the joint shall be repaired and subsequently radiographed. Repaired areas shall conform to the radiographic standard established for the particular joint.

5.6.3.5 Corresponding joints. On the next weldment designated for spot checking, the joint found rejectable will require spot checking only. Should a rejectable defect be found, the remainder of the joint shall be radiographed. All rejectable defects then found in the joint shall be repaired, and the repaired areas subsequently radiographed. Repaired areas shall conform to the radiographic standard established for the particular joint.

5.6.3.6 Checking of consecutive weldments. When a rejectable joint is found, the corresponding joint on the next weldment produced shall be completely radiographed. If no rejectable defects are found, spot checking will be resumed. Should rejectable defects be found, the corresponding joint on the weldment immediately following (in production) shall be completely radiographed. Complete radiography of the corresponding joint shall be continued, with each consecutive weldment produced, until a joint with no rejectable defects is obtained. All defects in each rejectable joint shall be repaired and the repaired areas shall conform to the radiographic standard established for the particular joint.

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5.6.3.7 Checking at intervals. When the total number of radiographic films required daily exceeds the maximum established in the contract, weldments shall be selected at intervals, rather than consecutively, in order not to interfere with the production schedule. These intervals shall be established by the Government and will be determined by the production schedule, the number of the joints for which the films are simultaneously required and the relative importance of the joints, as indicated by the frequency of spot checking required.

Custodian:
Army - AT

Preparing activity:
Army - AT

Review activity:
Army - MR

Project No. THJM-A088

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RECORDED JOINT WELDING PROCEDURE

FOR

MEDIUM TANK M1 (VEHICLE)

WELDED HULL (WELDMENT)

Book 10

DATE OF SUBMITTAL - 28 NOV 78

MANUFACTURED BY

Tank Manufacturing Company

LOCATED AT

Lincoln, Pennsylvania

FOR CONTRACTOR

ABC Corporation

LOCATED AT

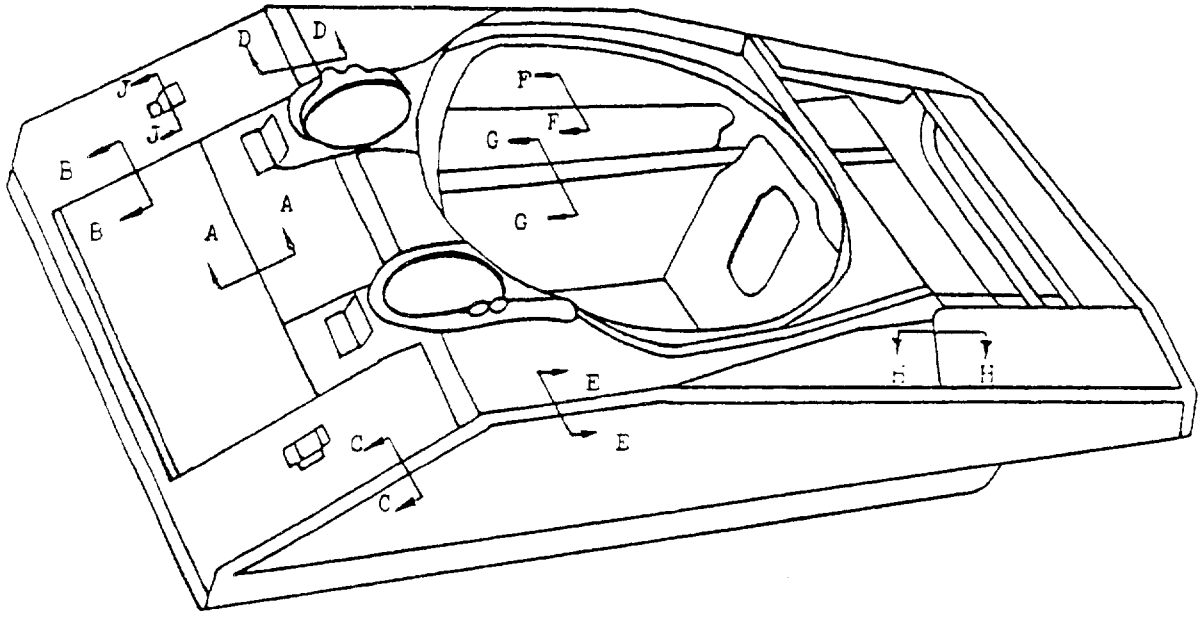
El Paso, California

Contract No.

Manufactured Under	Approved by Contractor Date
MIL-STD-XXX	Approved by Manufacturer Date

FIGURE 1. Example of cover page for recorded joint welding procedure.

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NOTES:

a. Joint details are shown in recorded joint welding procedures.

b. The joint designs shown in the recorded joint welding procedures are for illustration purposes only, and it is not necessarily indicated that applications of these joints in similar locations will be approved for any specific vehicle.

FIGURE 2. Example of perspective drawing showing location of weld joints in recorded joint welding procedure.

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Joint Section	Recorded joint welding procedure No.	Armor 1/ reference designation	Status of procedure certification					
			Ballistic tests			Workmanship specimens		
			Flat	Vertical	Horizontal	Flat	Vertical	Horizontal
A-A	1	I	Passed (specimen #17 & 18	Not tested	Not tested	Passed (specimen #50	Not required	Not required
B-B	2	I	Passed (specimen #17 & 18	Not tested	Not tested	Passed (specimen #50	Not required	Not required
		II	Passed (specimen #19	Not tested	Not tested	Passed (specimen #50	Not required	Not required
C-C	3	I	Passed (specimen #17 & 18	Not tested	Not tested	Passed (specimen #51	Not tested	Not tested
		II	Passed (specimen #22	Not tested	Not tested	Passed (specimen #51	Not tested	Not tested
D-D	4	I	Passed (specimen #16	Not tested	Not tested	Not required	Not required	Not required
		II	Passed (specimen #20 & 21	Not tested	Not tested	Not required	Not required	Not required
E-E	5	II	Passed (specimen #22	Not tested	Not tested	Passed (specimen #51	Not tested	Not tested
F-F	6	II	Passed (specimen #20 & 21	Not tested	Not tested	Passed (specimen #52	Not tested	Not tested
G-G	7	II	Passed (specimen #20 & 21	Not tested	Not tested	Passed (specimen #52	Not tested	Not tested

FIGURE 3. Example of summary of certification tests for recorded joint welding procedures.

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Joint section	Recorded joint welding procedure No.	Armor 1/ reference designation	Status of procedure certification					
			Ballistic tests			Workmanship specimens		
			Flat	Vertical	Horizontal	Flat	Vertical	Horizontal
H-H	8	II	Not required	Not required	Not required	Passed (specimen #53	Passed (specimen #54	Passed (specimen #55
J-J	9	I and II	Not required	Not required	Not required	Passed (specimen #53	Passed (specimen #54	Passed (specimen #55

1/ Summary of armor and filler data.

FIGURE 3. Example of summary of certification tests for recorded joint welding procedure. Continued

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Armor Plate Desig.	Type	Specification No.	Mfr.	Chemical composition range (percent) C	Mn	Si	S	P	Cr	Ni	Mo	V	Cu	Zr	Draw * Temp. °F(°C)
I	Rolled Homogeneous	MIL-A-12560	"X"	.26 .30	.95 1.15	.60 .90	.035 MAX	.035 MAX	.60 .70		.20 .25			.07 .11	800 (426.6)
II	Rolled Homogeneous	MIL-A-12560	"Y"	.23 .30	.85 1.00	.20 .30	.04 MAX	.03 MAX	.75 .90	.90 1.10	.30 .35		.07 .16		600 (315.5)

*Drawing temperature employed by the armor manufacturer in heat-treating armor.

FIGURE 4. Summary of armor types and compositions.

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Table A. Electrodes Conforming With Specification No.MIL-E-[illegible]

Electrode Reference Designation	Manufacturer	Brand Name	Type	Class
A	"X"	Bestweld	V	MIL-307L-15
B	"X"	Superweld	VI	MIL-308MOL-15
C	"Y"	Bestalloy	VI	MIL-308MOL-15
D	"Y"	Superloy	VI	MIL-308MOL-15
E	"Z"	Bestarc	V	MIL-307L-15
F	"Z"	Superarc	VI	MIL-308MOT-16

Table B. Electrodes Not Conforming With Specification No. MIL-E-[illegible]

Electrode Reference Designation	Manufacturer	Brand Name	Type of Covering	Chemical composition range (%)	C	Mn	Si	S	P	Cr	Ni	Mo	V	Others
[illegible]	"Q"	Excelweld	Lime	Core Wire	.15 Max.	3.50 4.50	.25 .60	.03 Max.	.04 Max.	19.5 21.5	9.0 10.5			
				Deposited Weld Metal	.17 Max.	5.00 4.50	.80 Max.			18.0 20.5		.50 1.00		
[illegible]	"R"	Wonderweld	Titania	Core Wire	.15 Max.	1.50 2.00	.25 .60	.03 Max.	.04 Max.	19.5 21.5	9.0 10.5			
				Deposited Weld Metal	.17 Max.	1.25 2.00	.80 Max.			18.0 20.5		1.80 2.25		

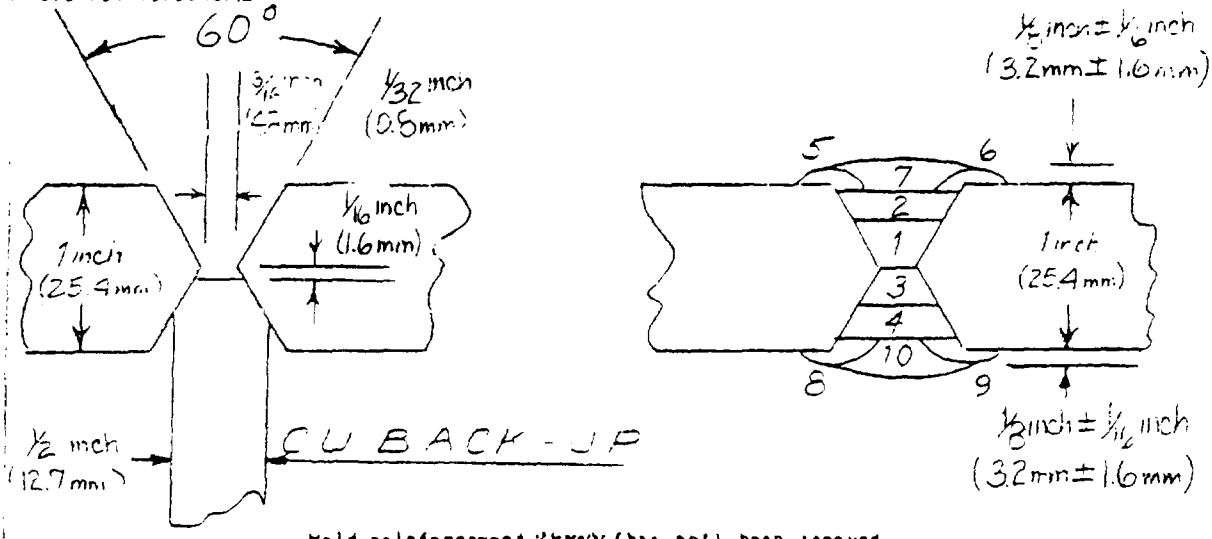
FIGURE 5. Summary of filler metal.

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WELDED ARMOR DATA

PLATE NO. 10X, 11X, 13X	SUBMITTED BY USATARCOM
DATE Jul, Aug 1976	
TYPE High Hard/Homo	ADDRESS 6501 11 Mile Rd
THICKNESS 1 inch (25.4 mm)	Warren, Michigan 48090
SPECIFICATION MIL-W-46086	CONTRACT NO.
ORDNANCE MATERIAL CONCERNED Steel Armored Vehicles	
WELDED BY A. Krzemecki	
OBJECT Development	

On a dimension sketch of the groove and weldment, indicate: (1) the included angle; (2) the root opening; (3) the root face; (4) the bead sequence; (5) additional sketch of spacer used on back-up, if any; (6) width of masking, if any, on edges of plate; (7) average height of weld reinforcement.



Weld reinforcement (has not) been removed.

WELDING DATA

PLATE PREPARATION: Flame Cutting

POSITION OF WELDING: Flat

WELDING: Automatic - Hand Semi POLARITY: - Rev - DC

PREHEAT 125F POSTHEAT None

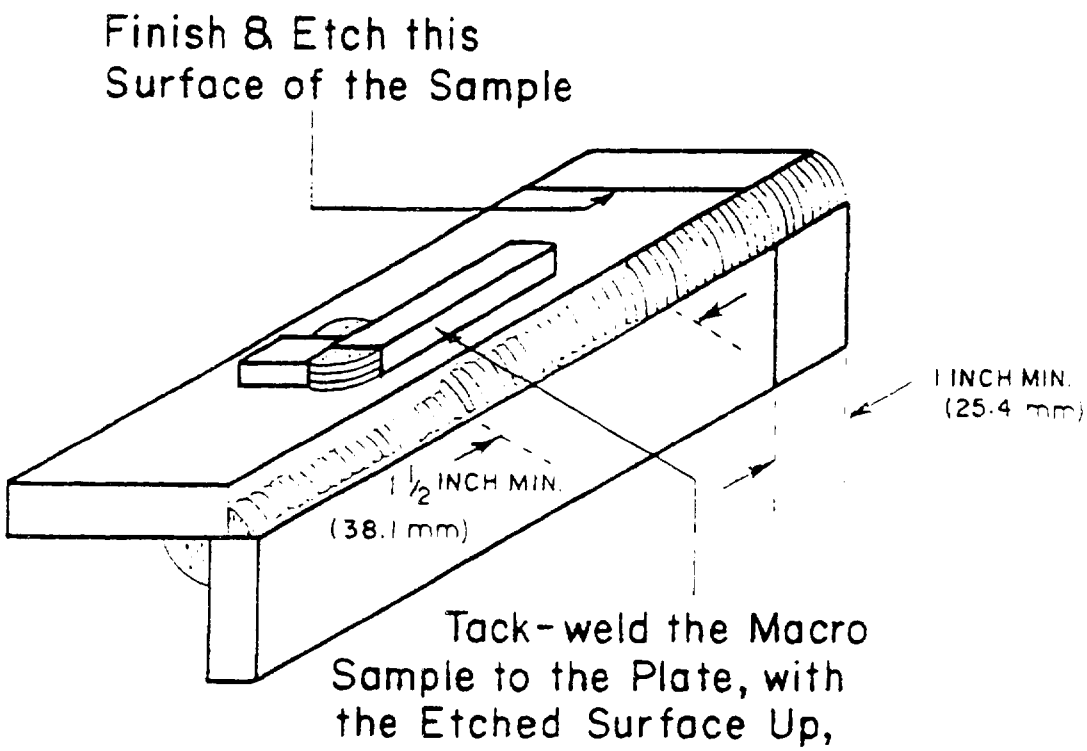
PEENING None BUTTERING None

PASS	ELEC SIZE (in/mm)	TYPE PASS	AMPS	VOLTS	CRACKING	CHIP OR GRIND	SPEED IN/. IN	INTERPASS TEMP °F(°C) A-B	B-C	C-D
1	.187/4.8	B	220	22	No	No		125°F(51.7°C)		
2	.062/1.9	W	270	24	"	"		"	"	"
3	" "	B	270	24	"	"		"	"	"
4	" "	W	270	24	"	"		"	"	"
5	" "	B	270	24	"	"		"	"	"
6	" "	B	270	24	"	"		"	"	"
7	" "	W	270	24	"	"		"	"	"
8	" "	B	270	24	"	"		"	"	"
9	" "	B	270	24	"	"		"	"	"
10	" "	W	270	24	"	"		"	"	"
11										
12										
13										
14										
15										
16										

B-BEADING PASS W-WEAVING PASS

FIGURE 5. Example of welded armor data for recorded joint welding procedure.

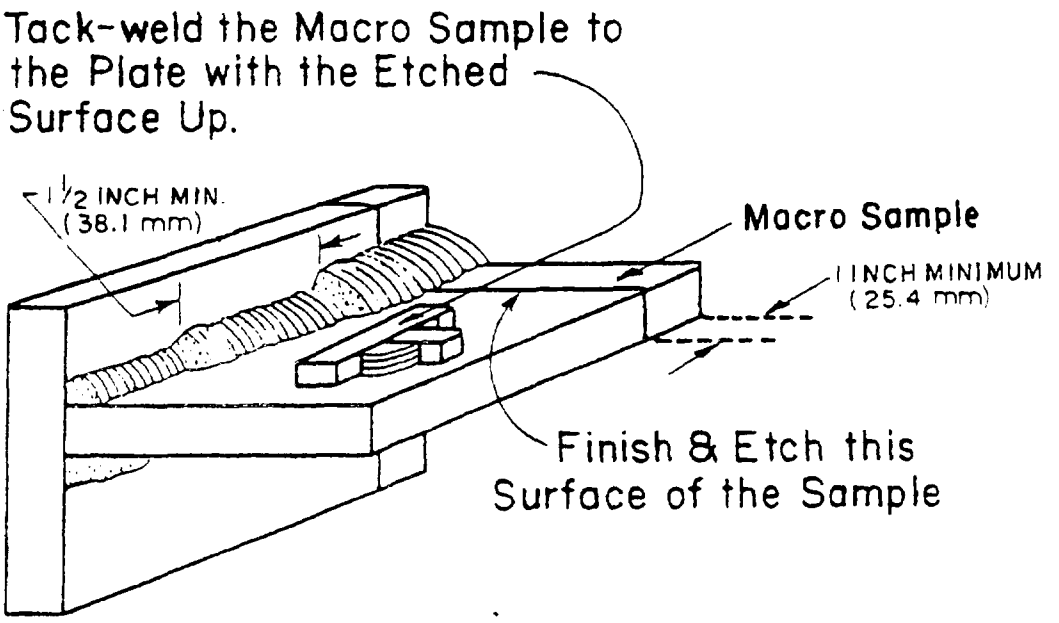
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Groove details of actual specimens shall conform to details of contractor's recorded joint welding procedure for each type of joint involved.

FIGURE 1. General form of workmanship specimen for corner joint type.

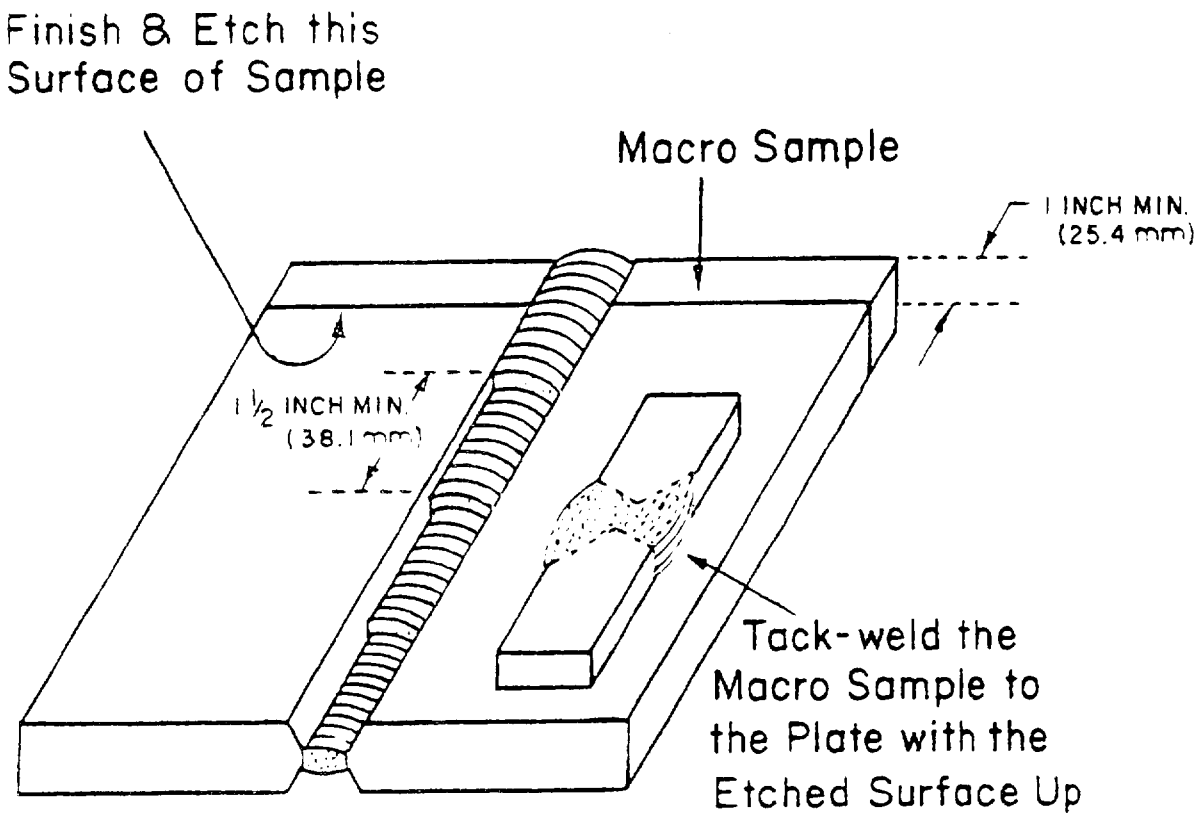
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Groove details of actual specimens shall conform to details of contractor's recorded joint welding procedure for each type joint involved.

FIGURE 8. General form of workmanship specimen for tee joint type.

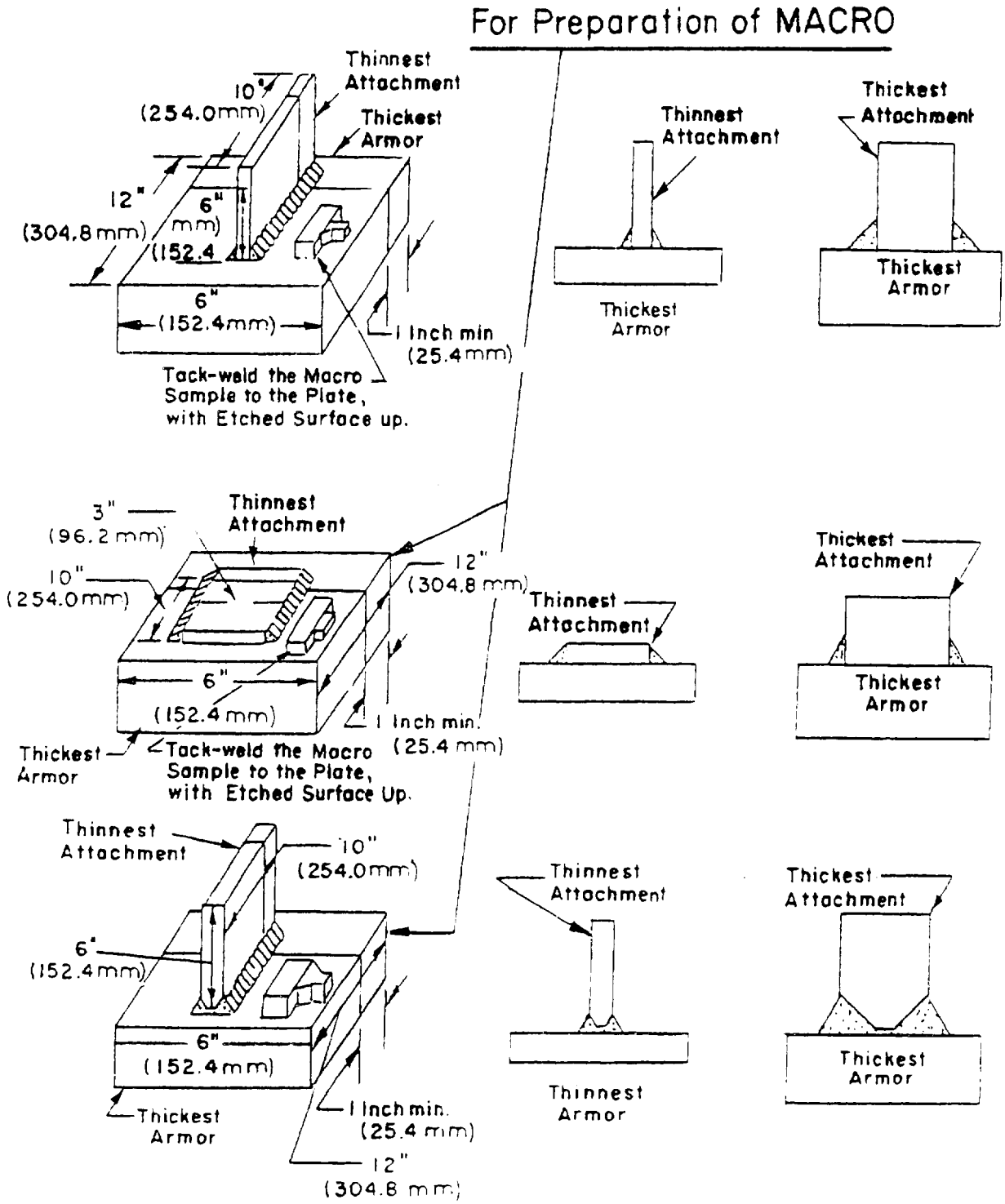
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Groove details of actual specimens shall conform to details of contractor's recorded joint welding procedure for each type of joint involved.

FIGURE 9. General form of workmanship specimen for butt joint type.

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Details of actual specimens shall conform to details of contractor's recorded joint welding procedure for each type of joint involved.

FIGURE 10. General form of workmanship for fillet welded attachment.

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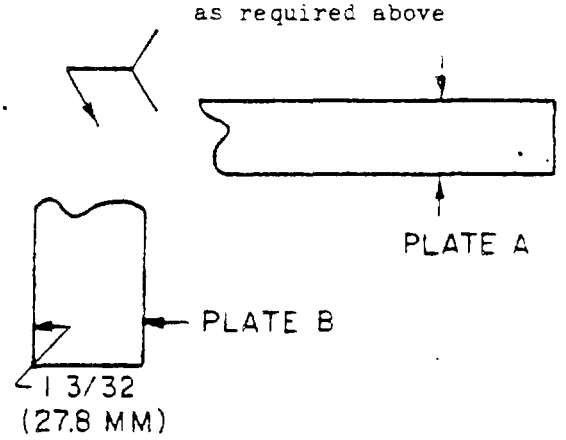
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Joint A-A

Plate No. 1
Type Filler Metal I
Thickness as Shown
Specification
Ordnance Material concerned: As shown
Welded by:
Object: Recorded Joint Welding Procedure No. 1 Joint A-A

On a dimensioned SKETCH of the Groove and Weldment, indicate (1) the included angle; (2) the root opening; (3) the root face; (4) the bead sequence; (5) additional sketch of spacer strip on back-up, if any; (6) width of masking, if any, on edges of plate; (7) average height of weld reinforcement.

Provide details of joint with dimensions as required above



RANGE OF CHEMICAL COMPOSITION FOR ARMOR

Plate	Spec	C	Mn	SI	S	P	CR	NI	MO	V	Cu	Zr
		.25	.95	.60	.035	.035	.60		.20			.07
A	MIL-A-12560	.30	1.15	.90	MAX	MAX	.70		.25			.11
		.23	.85	.20	.04	.03	.75	.90	.30		.07	
B	MIL-A-12560	.30	1.00	.30	MAX	MAX	.90	1.10	.35		.16	

FILLER METAL DATA

Size	Mfgr	Spec	Brandname	Designation	Type	Class
3/16 (4.76 mm)	"X"	12345	Bestweld	A	II	MIL-3071-15

FIGURE 11. Example of welded armor data for recorded joint welding procedure for ballistics test plates

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Plate preparation: Thermal Cutting - Sawing - Grinding - Machining
Plate Precleaning Method:

Position of Welding: Flat - Horizontal - Vertical - Overhead
Welding: Automatic - Semi-Automatic - Hand
Polarity: Str - Rev - AC - DC
Preheat:
Process
Type of Gas:
Power Source
Postheat:
Buttering:
Drive Unit

Pass	Elec size	Gas flow	Amperage range	Voltage range	Nozzle I.D.	Chip or grind	Wire speed in/min	Interpass Temp. °F (°C)
1								
2								
3								
4								
5								
6								
7								
8								
9								
10								

FIGURE 12. Example of welding data sheet for ballistics test plates.

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(1) REPORT NO.				SHEET NO	OF
(2) X-RAY SERIAL NO.					
(3) PLATE SUBMITTED BY			(4) PLATE NO.		(5) SPEC.
(6) RADIOGRAPHED BY					(7) DATE
(8) PLATE THICKNESS	(9) KV	(10) MA	(11) TIME		(12) FOCAL DIST
(13) TYPE OF FILM			(14) SCREENS OR FILTER		

SHOCK TEST SPECIMEN

Showing Locations of Radiographs and Results of Test

(15) CRACK — INCOMPLETE FUSION — INCOMPLETE PENETRATION

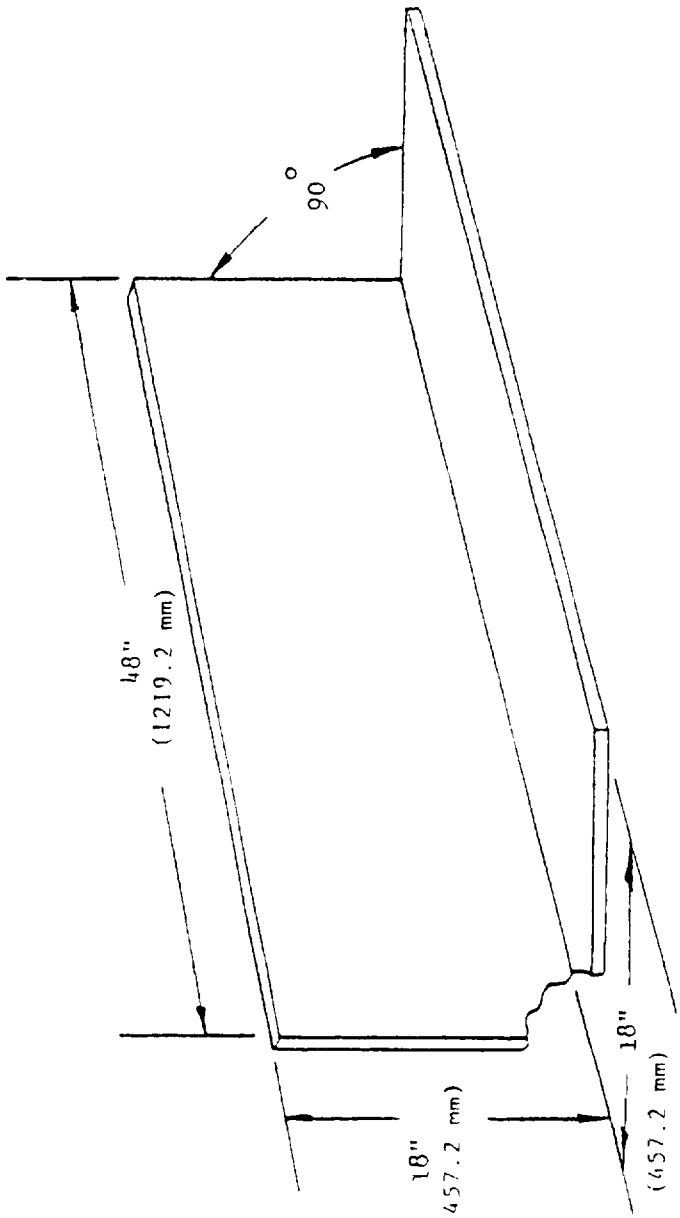
POROSITY AND SLAG INCLUSIONS — UNDERCUTTING

(16) RESULTS

(17) NEGATIVES READ BY

FIGURE 13. Example of weld radiographic report for ballistics test plates.

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NOTE: The 90° angle may vary to reflect condition in vehicle

FIGURE 14. Drawing of ballistic corner joint specimen.

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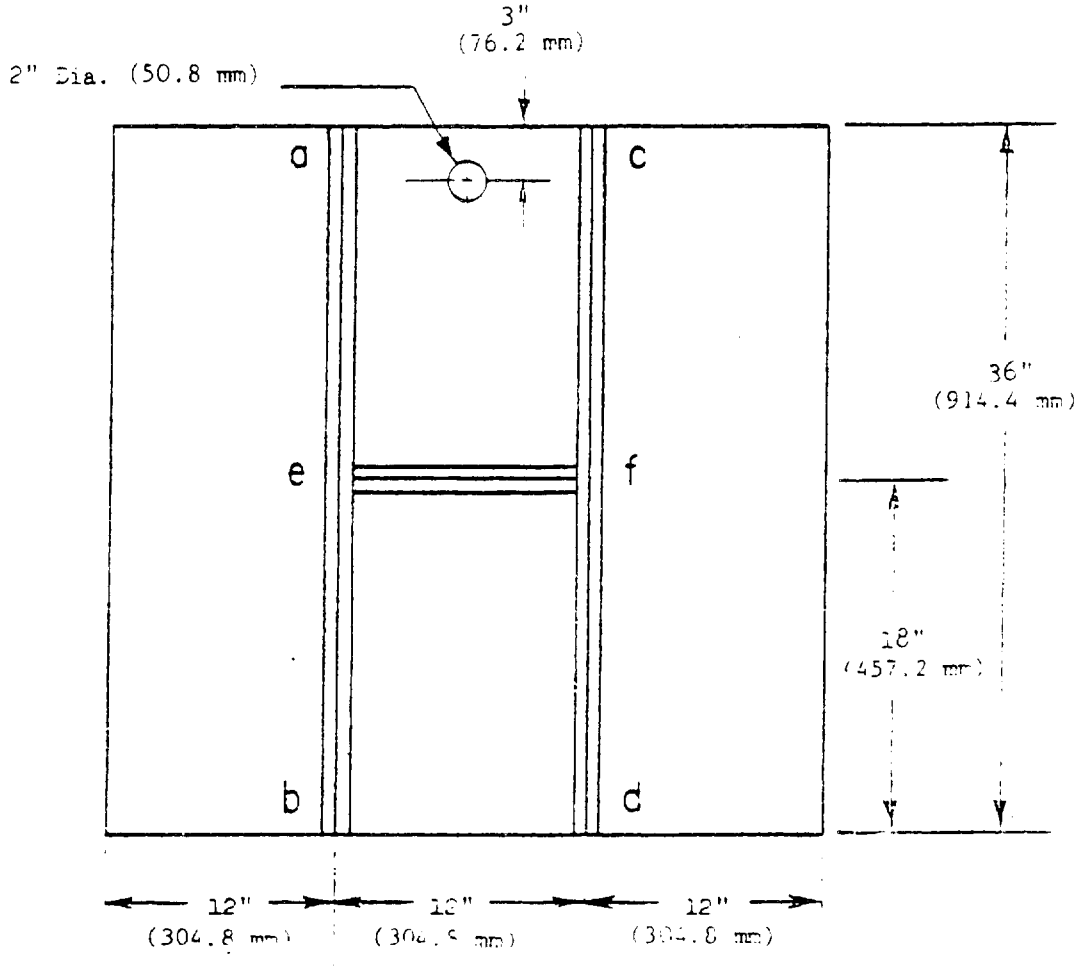


FIGURE 1. Drawing of ballistic test (H) plate.

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